



National Oceanic and
Atmospheric Administration
U.S. Department of Commerce

Lightning

Introduction to Lightning

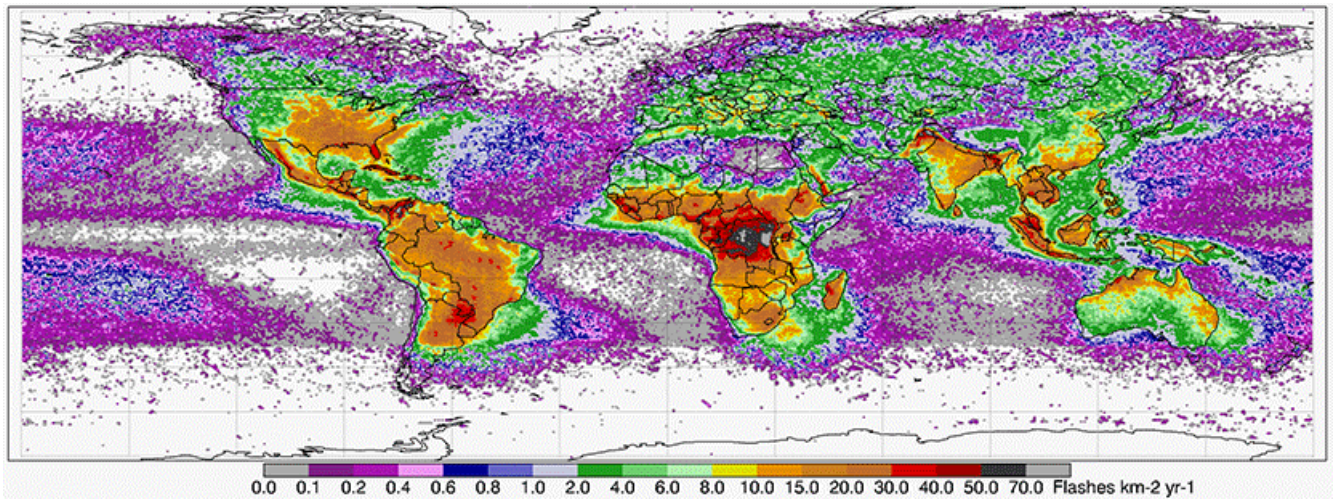
While lightning is simply a gigantic spark of static electricity (the same kind of electricity that sometimes shocks you when you touch a doorknob), scientists do not have a complete grasp on how it works.

Lightning has been seen in volcanic eruptions, extremely intense forest fires, surface nuclear detonations, heavy snowstorms, and large hurricanes. However, it is most often seen in thunderstorms. In fact, lightning (and the resulting thunder) is what makes a storm a thunderstorm.

At any given moment, there can be as many as 2,000 thunderstorms occurring across the globe. This translates to more than 14.5 MILLION storms each year. NASA satellite research indicates that these storms produce lightning flashes about 40 times a second worldwide.

This is a change from the previously-accepted value of 100 flashes per second, which was estimated in 1925. Whether it is 40, 100, or somewhere in between, we live on an electrified planet.

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Annual number of lightning flashes based on observations from NASA satellites - From [High Resolution Full Climatology...](#)

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